Technology Stack (Architecture & Stack)

Date	2 October 2026
Team ID	6abca5b16f395c49751f124d
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	4 Marks

Technical Architecture

LegalEase uses a separated frontend/backend architecture. Streamlit collects user inputs and presents the editable preview. FastAPI validates the request and routes it to the Gemini generation service. Export utilities convert the latest edited content into TXT, DOCX and PDF.

S.No	Component	Description	Technology
1	Frontend	Form input, preview, editing and downloads	Streamlit
2	Backend	REST API, request validation and orchestration	Python, FastAPI, Uvicorn
3	AI Core	Legal-document draft generation	Google GenAI SDK, current Gemini model
4	Validation	Typed API request validation	Pydantic
5	DOCX Export	Formatted Microsoft Word output	python-docx
6	PDF Export	Multi-page formatted PDF output	ReportLab
7	TXT Export	UTF-8 plain-text output	Python
8	Configuration	Environment-based API key/model settings	python-dotenv
9	Testing	Backend and export tests	pytest

Application Characteristics

Characteristic	Requirement
Modularity	Separate frontend, API, AI and export modules
Security	API key excluded from source control
Usability	Guided fields, preview, editing and downloads
Safety	Clear educational/informational disclaimer; no guarantee of legal validity

Portability	Runs locally with Python 3.11+ and a web browser